

CC LAB-06 - Jenkins and Nginx

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SECTION : F

Task 1: Set Up Jenkins Using Docker

SS1

The screenshot shows the Jenkins 'Getting Started' page in a web browser. The browser's address bar displays 'localhost:8080'. The page features a progress bar and a table of installed and available plugins. Below the table, there is a list of installed plugins categorized by type (Instance Identity, Mailer, Pipeline, Git, GitHub, etc.). The Jenkins version '2.541.2' is displayed at the bottom.

Getting Started			
Getting Started			
✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding
✓ Timestampers	✓ Workspace Cleanup	✓ Ant	✓ Gradle
✓ Pipeline	✓ GitHub Branch Source	✓ Pipeline: GitHub Groovy Libraries	⌚ Pipeline Graph View
✓ Git	⌚ SSH Build Agents	⌚ Matrix Authorization Strategy	⌚ LDAP
⌚ Email Extension	✓ Mailer	⌚ Dark Theme	

** Instance Identity

Mailer

- ** Branch API
- ** Pipeline: Multibranch
- ** Pipeline: Stage Tags Metadata
- ** Pipeline: Input Step
- ** Pipeline: Basic Steps
- ** Pipeline: Declarative

Pipeline

- ** Java JSON Web Token (JJWT)
- ** GitHub API
- ** Mina SSHD API :: Common
- ** Mina SSHD API :: Core
- ** Gson API
- ** Git client

Git

- ** GitHub

GitHub Branch Source

Pipeline: GitHub Groovy Libraries

** - required dependency

Jenkins 2.541.2

SS2

The screenshot shows the Jenkins web interface in a browser window. The address bar displays 'localhost:8080'. The Jenkins logo and name are at the top left. A user menu is open on the top right, showing the user 'pes2ug23cs391' and options: Theme (set to Light), My Views, Account, Appearance, Preferences, Security, Experiments, Credentials, and Sign out. The main dashboard includes a 'Build Queue' section with 'No builds in the queue.', a 'Build Executor Status' section, and a 'Welcome to Jenkins!' message. Below the welcome message, it states: 'This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.' There are two main sections: 'Start building your software project' with a 'Create a job' button, and 'Set up a distributed build' with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The footer shows the URL 'localhost:8080/user/pes2ug23cs391', 'REST API', and 'Jenkins 2.541.2'.

localhost:8080

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

localhost:8080/user/pes2ug23cs391

REST API Jenkins 2.541.2

Task-2: Create a Jenkins Job to Build the Backend

SS3



```
#6 93.58 Setting up libtiff5:amd64 (4.3.0-6ubuntu0.12) ...
#6 93.60 Setting up libfontconfig1:amd64 (2.13.1-4.2ubuntu5) ...
#6 93.62 Setting up binutils-x86-64-linux-gnu (2.38-4ubuntu2.12) ...
#6 93.64 Setting up libxpm4:amd64 (1:3.5.12-1ubuntu0.22.04.2) ...
#6 93.66 Setting up binutils (2.38-4ubuntu2.12) ...
#6 93.68 Setting up libgd3:amd64 (2.3.0-2ubuntu2.3) ...
#6 93.70 Setting up libstdc++-11-dev:amd64 (11.4.0-1ubuntu1~22.04.3) ...
#6 93.72 Setting up gcc-11 (11.4.0-1ubuntu1~22.04.3) ...
#6 93.74 Setting up libc-devtools (2.35-0ubuntu3.13) ...
#6 93.76 Setting up g++-11 (11.4.0-1ubuntu1~22.04.3) ...
#6 93.78 Setting up gcc (4:11.2.0-1ubuntu1) ...
#6 93.83 Setting up g++ (4:11.2.0-1ubuntu1) ...
#6 93.88 update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
#6 93.89 update-alternatives: warning: skip creation of /usr/share/man/man1/c++.1.gz because
associated file /usr/share/man/man1/g++.1.gz (of link group c++) doesn't exist
#6 93.90 Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
#6 DONE 94.7s

#7 [3/5] WORKDIR /app
#7 DONE 0.2s

#8 [4/5] COPY app.cpp .
#8 DONE 0.1s

#9 [5/5] RUN g++ app.cpp -o app
#9 DONE 1.2s

#10 exporting to image
#10 exporting layers
#10 exporting layers 2.1s done
#10 writing image sha256:7dd6aad305991139c44c1c68205274ea0ed4492eb05e684ff7583815b104fcb6 done
#10 naming to docker.io/library/backend-app done
#10 DONE 2.1s
Finished: SUCCESS
```

← → ↻ 🏠 localhost:8080/job/pes2ug23cs391-backend-build/ ☆ 🔍 ⚙️ 👤

Jenkins / pes2ug23cs391-backend-...

Status ✓ **pes2ug23cs391-backend-build** 📄 Add description

</> Changes

📁 Workspace

▶ Build Now

⚙️ Configure

🗑️ Delete Project

📄 Git Polling Log

✎ Rename

Permalinks

- Last build (#4), 3 min 14 sec ago
- Last stable build (#4), 3 min 14 sec ago
- Last successful build (#4), 3 min 14 sec ago
- Last failed build (#3), 6 min 31 sec ago
- Last unsuccessful build (#3), 6 min 31 sec ago
- Last completed build (#4), 3 min 14 sec ago

Builds > ...

🔍 Filter /

Today

✓ #4 7:04 AM

Task-3: Parameterized Jenkins Job

SS5

← → ↻ 🏠 localhost:8080/job/pes2ug23cs391-backend-build/build?delay=0sec ☆ 🔍 ⚙️ 👤

Jenkins / pes2ug23cs391-backend-... ▾

Project pes2ug23cs391-backend-build

This build requires parameters:

Backend_Count

1 ▾

▶ Build Cancel

Builds > ...

🔍 Filter /

Today

✓ #5 7:13 AM ▾

✓ #4 7:04 AM ▾

SS6

✓ / #5 ✓



```
#4 [1/5] FROM docker.io/library/
ubuntu:22.04@sha256:3ba65aa20f86a0fad9df2b2c259c613df006b2e6d0bfcc8a146afb8c525a9751
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 29B done
#5 DONE 0.0s

#6 [2/5] RUN apt-get update && apt-get install -y g++
#6 CACHED

#7 [3/5] WORKDIR /app
#7 CACHED

#8 [4/5] COPY app.cpp .
#8 CACHED

#9 [5/5] RUN g++ app.cpp -o app
#9 CACHED

#10 exporting to image
#10 exporting layers done
#10 writing image sha256:7dd6aad305991139c44c1c68205274ea0ed4492eb05e684ff7583815b104fcb6 done
#10 naming to docker.io/library/backend-app done
#10 DONE 0.0s
+ docker rm -f backend1 backend2
Error response from daemon: No such container: backend1
Error response from daemon: No such container: backend2
+ [ = 1 ]
+ docker run -d --name backend1 backend-app
c02d6adda43930af603c470d71db58521108a7c11709d1b85d2686bbe41bc419
+ docker run -d --name backend2 backend-app
bfe6688144260d406b9c4d17ddd6972811dedf2fd329942a88bf081fca2e5c2
Finished: SUCCESS
```

▼ / #6 ▼

🔍

⚙️

👤

ubuntu:22.04@sha256:3dab3aa20t8ba0Tad9dT2D2C259C013dT000D2E000DTC08A140ATD8C525A9/51

#4 DONE 0.0s

#5 [internal] load build context

#5 transferring context: 29B done

#5 DONE 0.1s

#6 [2/5] RUN apt-get update && apt-get install -y g++

#6 CACHED

#7 [3/5] WORKDIR /app

#7 CACHED

#8 [4/5] COPY app.cpp .

#8 CACHED

#9 [5/5] RUN g++ app.cpp -o app

#9 CACHED

#10 exporting to image

#10 exporting layers done

#10 writing image sha256:7dd6aad305991139c44c1c68205274ea0ed4492eb05e684ff7583815b104fcb6 done

#10 naming to docker.io/library/backend-app

#10 naming to docker.io/library/backend-app done

#10 DONE 0.0s

+ docker rm -f backend1 backend2

backend1

backend2

+ [= 1]

+ docker run -d --name backend1 backend-app

4c6973c76bc544752455622e6224337e61ea5ca28692c3a185f8bd812d58af14

+ docker run -d --name backend2 backend-app

8a1d07d68a2f0bf60ac52a6a94fd3d4526242cee8f06bb45dbf39ca3b4f278a6

Finished: SUCCESS

TASK-4 — Jenkins Pipeline for Automated Deployment

SS7

[Rerun](#) [Settings](#)

Graph

```
graph LR; Start((Start)) --> Checkout[Checkout SCM]; Checkout --> Build[Build Backend Image]; Build --> DeployBackend[Deploy Backend...]; DeployBackend --> DeployNGINX[Deploy NGINX Loa...]; DeployNGINX --> PostActions[Post Actions]; PostActions --> End((End));
```

Start Checkout SCM Build Backend Image Deploy Backend... Deploy NGINX Loa... Post Actions End

Post Actions 73ms Started 3m ago Jenkins

✓ Pipeline executed successfully. NGINX load balancer is running. 33ms



Jenkins

LAB6-PIPELINE-NGINX

#3



```
f73400a233fd: Verifying Checksum
f73400a233fd: Download complete
47cd406a84ef: Verifying Checksum
47cd406a84ef: Download complete
bae5a1799a80: Verifying Checksum
bae5a1799a80: Download complete
0c8d55a45c0d: Pull complete
46bf3a120c8e: Verifying Checksum
46bf3a120c8e: Download complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
a2cd9babe15f12b58a693af4d564aeb23bab74ab5e49ad53c025c6d2d8a10b7
+ docker cp nginx/default.conf nginx-lb:/etc/nginx/conf.d/default.conf
+ docker exec nginx-lb nginx -s reload
2026/02/19 07:42:33 [notice] 30#30: signal process started
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully. NGINX load balancer is running.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

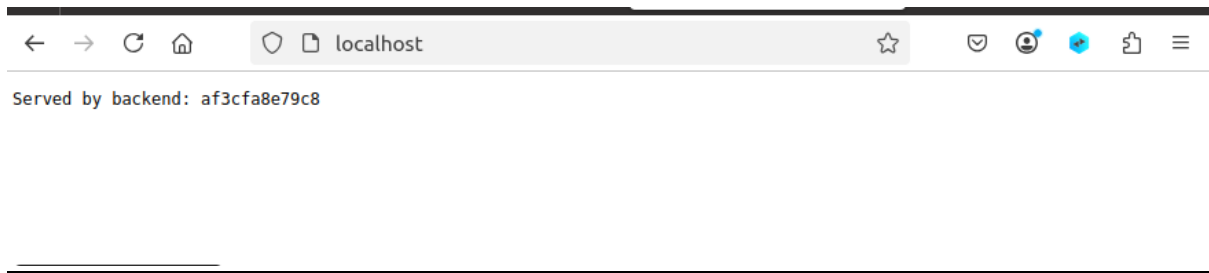
SS9



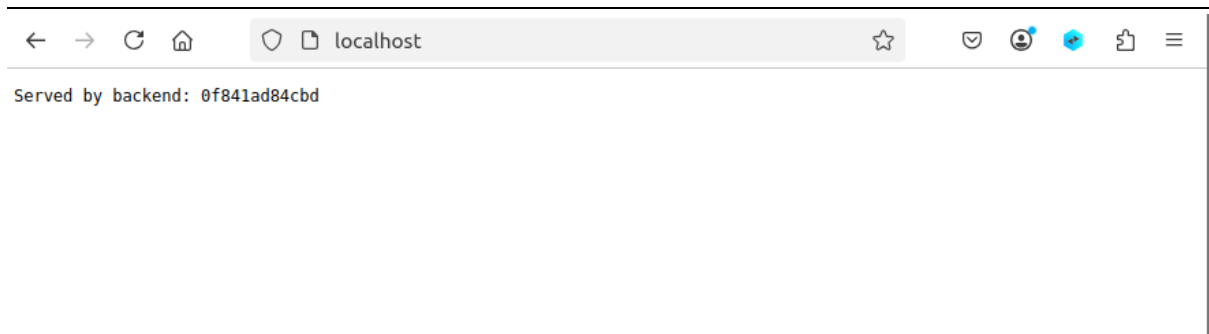
localhost



Served by backend: be160827919b



Task-5: NGINX load-balancing strategies



Same backend id always

